Muaz Iqubal

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SUMMARY

Third-year Computer Science undergraduate (Class of 2028) who builds and ships developer tools in Python: a static analyser for LLM agent costs, published on PyPI and evaluated on 1.9M lines of open-source code; an automated job-alert system running hourly in CI; and long-running desktop apps. Seeking a Summer 2027 SWE internship.

EDUCATION

PES University, Bangalore *Expected Graduation: 2028*
B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning)
Relevant coursework: Data Structures, Algorithms, Object-Oriented Programming, System Design

TECHNICAL SKILLS

Languages: Python, JavaScript, C
CS Fundamentals: Data Structures, Algorithms, System Design, Static Analysis (ASTs, call graphs)
Libraries & Frameworks: FastAPI, SQLAlchemy, React, PySide6 (Qt), Pillow, OpenCV, Pytesseract, Tkinter
AI & LLM Tooling: LLM SDK call patterns (OpenAI, Anthropic, Gemini, LiteLLM, LangChain), Ollama, Model Context Protocol (MCP), REST and OAuth APIs
Developer Tools: Git, GitHub Actions (CI/CD), pytest, mypy, Ruff, PyPI packaging, PostgreSQL, SQLite, Vercel

PROJECTS

AgentGuard — Static Analyser for LLM Agent Costs *github.com/muaz-q/AgentGuard*
- Built an AST-based analyser that finds what makes LLM agents expensive — uncapped loops, history re-sent every turn (quadratic input tokens), uncapped outputs — following calls across functions and files.
- Measured **92% precision** by hand-checking every finding on a held-out set of 8 open-source agent projects (1.1M lines), after tuning rules on a separate 14; it flagged unbounded history growth in OpenAI's swarm.
- Validated the cost model on real LLM runs: **within 11%** of measured tokens at 40 turns (naive: 99% low).
- Shipped as a CLI, GitHub Action (SARIF code-scanning alerts), pre-commit hook and MCP server for AI coding agents; 136 tests, CI on Linux, Windows and macOS, released to PyPI as *agentguard-lint*.

Job Radar — Automated Internship Alert System *github.com/muaz-q/job-radar*
- Built an hourly monitor for 29 company careers sites (Workday, Greenhouse, Lever, Ashby, Amazon, Microsoft), Unstop and Wellfound that normalises every listing and sends Telegram alerts for new matching roles.
- Designed three-level deduplication (posting id, content key, cross-site reworded-title match); state is saved before alerts are sent, so a failed run can neither lose nor repeat a job.
- Cut scan time 5.6× (288 s to 51 s) with parallel, per-host rate-limited fetching; deployed on GitHub Actions and Vercel with 233 automated test cases run in CI against recorded real API responses.

CS Geek — Daily Trivia Game with Measured Answer Popularity *stack-rank-one.vercel.app*
- Built and deployed a daily trivia game (inspired by Krillion) where rarer correct answers score higher; popularity is measured from real players in Postgres (Vercel serverless + Neon), one vote per player per question.
- Seeded 29 questions from published data (Stack Overflow Developer Survey, StatCounter, JetBrains); seeds blend into live player counts as data arrives, and every other question is labelled in-game as an estimate.
- Wrote typo-tolerant answer matching (edit distance, aliases, hints) over 526 questions and ~23,000 accepted answers, and log rejected guesses to find missing answers; mobile-first canvas UI shared by two editions.

Sally — Real-Time Spotify Wallpaper Engine *github.com/muaz-q/Sally*
- Built a Windows system-tray app that runs 24/7, rebuilding the desktop wallpaper from album art whenever the Spotify track changes and restoring the user's own wallpaper on exit (PySide6, Pillow, Win32 API, OAuth).
- Timed each Spotify API check to the end of the current track, so song changes appear within about 1 s and skips within 5 s while staying clear of the API's rate limits.

EXPERIENCE

Outreach & Marketing Intern — KaizenDev
- Ran outreach campaigns in developer communities to drive product awareness and user growth, translating technical product detail into developer-facing messaging.

ADDITIONAL

Writing: *Living or Existing?* — a collection of ten personal essays on purpose, loneliness, failure and loss (2026); more essays on existentialism and philosophy at medium.com/@mwahz | **Certification:** UI/UX Design (Udemy)